

Highlight of Skills

- 3+ years research and industry experiences in data analysis to extract insights and support data-driven decision-making across multidisciplinary projects.
- Experience working within a Linux environment and HPC environment (supercomputer).
- Proficient in communicating analytical findings through data visualizations and stakeholder oriented reports.
- Experiences working in collaborative teams, strong problem-solving and critical thinking skills, attentive to details.
- Languages: English, Cantonese, Mandarin
- Tools: SQL, R, Python, Excel, Tableau, Power BI, Git, QGIS

Education

- Sep 2025 – **MSc in Statistics**, *University of Toronto*, Toronto, ON, Canada
Apr 2026 (expected) Courses: Applied Multivariate Analysis, Methods of Applied Statistics, Special Topics in Statistical Data Science, Special Topics in Applied Statistics, Survival Analysis
- Sep 2021 – **H.B.Sc in Statistical Science (Specialist), Mathematics (Major), Health Studies (Focus)**, *University of Toronto*, Toronto, ON, Canada **cGPA: 3.94/4.00**
Apr 2025 Courses: Applied Statistical Methods, Applied Spatial Analysis, Time Series Analysis, Surveys, Sampling and Observational Data, Statistical Consultation
Awards: Samuel Beatty In-course Scholarship (2025), University of Toronto Excellence Award (2024), Eleventh Annual Canadian Statistics Student Conference Undergraduate Level First Prize (2023), University of Toronto Dean's List Scholar (2021–2025)

Data-related Experiences

- Sep 2025 – **R-Based Learning Designer**, *University of Toronto Human Biology Program*
Current
 - Developed 4 auto-testable R coding assignments for a cohort of 400 students, improving pedagogical effectiveness and reduced instructor grading workload.
 - Designed 2 interactive LearnR modules with guided coding exercises and visualizations using RShiny, improving skills in reproducible coding, user-centered design, and data visualization.
 - Collaborated with peer module developers to address issues and revise updated modules, strengthening teamwork and technical communication skills.
- Sep 2025 – **Statistical Neuroimaging Researcher**, *University of Toronto, Department of Statistical Sciences*
Dec 2025
 - Investigated statistical inference for genetic heritability of intermodal coupling using simulated multi-modal brain MRI data, providing insights for advanced methodological approaches in neuroimaging data.
 - Conducted 50+ large-scale simulation experiments in HPC environment and applied adaptive clustering techniques with parallel computing, validating proposed model's ability in controlling Type I error rates and statistical power.
 - Summarized findings in a detailed report, providing evidence that informed refinement of current methods and shaped the development of more robust neuroimaging statistical approaches.
- May 2024 – **Regression Analysis Researcher**, *University of Toronto Department of Statistical Sciences*
– Current
 - Conducted research on robust regression methods for handling outliers and heavy-tailed data, resulting in a manuscript in preparation and a department research day presentation.
 - Developed and evaluated 4 parameter estimation approaches under high-dimensional and outlier settings, providing insights for improving traditional estimation accuracy in different settings.
 - Implemented and troubleshoot optimization algorithms to address non-convergence issues and enhanced computational efficiency, enhancing proficiency for statistical computing through utilizing software R and Python.
 - Presented research findings through detailed reports and published a R package with 20+ reusable functions, strengthening statistical communication and facilitate research reproducibility.

- Sep 2024 – **Machine Learning Researcher**, *University of Toronto Department of Statistical Sciences*
- Apr 2025
- Preprocessed EHR data from 40,000+ ICU patients in the MIMIC-III database by joining relational tables, aggregating diagnosis codes, and standardizing clinical concepts for downstream machine-learning fairness evaluation.
 - Evaluated the SVD-PMI embeddings-building algorithm to assess its ability to capture semantic relationships among 6,000+ medical concepts, and conducted feature selection to optimize downstream predictive model performance.
 - Trained regression and random forest models for depression diagnosis and conducted fairness evaluations across demographic groups, revealing disparities in prediction outcomes linked to biased label distributions.
 - Maintained a reproducible workflow in R and curated a GitHub repository to share code and results, strengthening transparency and enabling replication by other researchers.
- Sep 2024 – **Data Visualization Analyst**, *University of Toronto Human Biology Program*
- Apr 2025
- Analyzed relationships between student engagement with lecture videos and discussion boards in Quercus and final course grades using Excel, guiding instructors in redesigning assessments and reallocating instructional resources.
 - Developed 100+ Tableau dashboards (cohort performance summaries, engagement piecharts, grade distribution boxplots etc), showcasing data visualization proficiency and strong statistical communication skills.
 - Collaborated with peers to troubleshoot data-quality issues and refined visualization approaches, strengthening problem-solving and teamwork skills.
- Jun 2024 – **Psychiatric Education Quantitative Researcher**, *Centre for Addiction and Mental Health*
- Apr 2025
- Examined how physical movement influences cybersickness symptoms across two immersive VR medical training modules, contributing to a peer-reviewed publication and two conference poster presentations.
 - Curated and validated survey data by checking for invalid or inconsistent entries, and restructuring subcategory scoring to avoid double-counting, ensuring a clean, reliable dataset for downstream analysis.
 - Summarized descriptive statistics and conducted non-parametric statistical analyses (Mann-Whitney U-tests) in R, identifying a significant positive association between movement intensity and nausea severity.
 - Provided advices for VR training programs design, highlighting movement thresholds and comfort considerations to help instructional teams balance user well-being with educational effectiveness.
- Sep 2023 – **Agent-Based Modeling Researcher**, *University of Toronto COBWEB lab*
- Apr 2024
- Initiated research using agent-based simulation to investigate declining birth rates and potential solutions, resulting in a poster presentation.
 - Conducted comprehensive literature reviews and prepared an annotated bibliography of 30+ sources, synthesizing multidisciplinary findings that improved study design and strengthened critical thinking and organizational skills.
 - Mastered the simulation software COBWEB to model population dynamics and policy scenarios, independently developing 20+ simulations that expanded research scope and demonstrated advanced self-learning ability.
 - Performed time-series and regression analyses in Python to assess the impact of policy interventions on birth rates trends, providing insights for short-term and long-term strategic proposal.
- May 2023 **Data Analyst Intern**, *Canadian Urban Institute*
- Aug 2023
- Developed regression models in R to analyze nationwide household- and business-level carbon emissions, and social connection patterns, providing data-driven insights for an ongoing Canadian city planning sustainability project.
 - Created an interactive tool to demonstrate the effects of urbanization and housing composition on greenhouse gas emissions, enabling stakeholders to model scenarios for policy planning.
 - Utilized geospatial visualization tool QGIS to produce 300+ detailed maps, combined with statistical findings, revealing urban planning insights and actionable recommendations for decision-makers.
- Oct 2022 – **Computational Social Science Researcher**, *University of Toronto Population Well-being Lab*
- May 2023
- Applied machine learning methods (Gaussian Mixture Model, k-means, and hierarchical clustering) to survey data consisting of 150K+ observations in R, to uncover meaningful subpopulations based on life satisfaction and demographics.
 - Prepared detailed 4 tip-sheets on algorithms and evaluated effectiveness and appropriateness of statistical methods, enhancing team understanding in methodological application.
 - Designed a conference poster discussing the suitability of clustering methods and implications of results, effectively explaining complex workflow for diverse audiences, resulting in a conference poster first prize.
- Sep 2022 – **Statistical Consultant**, *University of Toronto Data Sciences Café*
- Dec 2023
- Contributed to a supportive and collaborative consulting community, fostering an environment where interdisciplinary research and statistical analysis could thrive.
 - Recommended tailored statistical methodologies and identified limitations in proposed study designs to 20+ clients, delivering actionable insights that improved methodological validity and analytical outcomes across diverse fields.
 - Identified potential biases and limitations in proposed research methods, providing actionable insights for improvement in study method design.
 - Prepared 5 reports on advanced statistical topics, introducing statistical concepts and coding methods in an accessible manner to enhance statistical literacy among non-statistician researchers.

Statistics Course Projects

- Oct 2025 – **Uncovering Calendar-Based Structures Through Clustering of Daily Traffic Volume Profiles**, STA2005
Dec 2025
- Applied PCA and UMAP dimensionality reduction techniques in R to year-long traffic volume data across 26 locations, uncovering recurring temporal clusters tied to school calendars, holidays, and seasonal routines.
 - Created visualizations and prepared written report to communicating traffic flow patterns to non-technical stakeholders, delivering actionable insights for transportation planning and infrastructure maintenance.
- Oct 2025 – **Fatal US School Shootings (1999-2025)**, STA4101
Dec 2025
- Modeled fatality risk in 400+ US school shootings using logistic regression, highlighting actionable risk indicators including shooting intent, absence of non-shooter injuries, and shooter connection to the school community.
 - Developed a reproducible GitHub repository with version control documenting all data cleaning, modeling, and analysis steps for transparency and collaboration.
 - Prepared a written report synthesizing statistical findings into clear, evidence-based insights, informing safety and emergency preparedness strategies.
- Oct 2024 – **Transit and Crime: Investigating Assault Clusters Around Toronto's TTC Subway Lines**, STA465
Dec 2024
- Conducted spatial analysis of 200k+ Toronto assault incidents using kernel density estimation, point process models and HDBSCAN clustering, successfully identified assault concentration patterns within varying buffer distances around TTC subway infrastructure.
 - Developed interactive geospatial report with maps translating statistical findings into actionable insights for urban planners and policymakers on transit-related public safety.
- Sep 2024 – **Quercus Engagement as a Predictor of First-Year University Student Dropout Risk: A Logistic Regression Analysis**, STA490
Apr 2025
- Leveraged R to preprocess and analyze engagement data from 3,000+ first-year students, improving data quality by resolving missing value issues and standardizing predictors across multiple time windows.
 - Built and validated logistic regression models to assess the relationship between Quercus engagement and dropout risk, identifying early disengagement in first 4 weeks as a critical retention indicator.
 - Summarized findings in stakeholder-focused reports with clear visualizations and accessible interpretations, providing foundational evidences which informed refinement of current learning support strategies.
- Feb 2025 – **Modeling Cocoa Futures Prices: A Statistical Analysis of Ghana's Cocoa Production**, STA457
Apr 2025
- Integrated four large-scale datasets with 3,700+ daily observations and performed comprehensive data cleaning, including missing-value imputation, correction of invalid or contradictory entries, and monthly aggregation, to produce a clean, analysis-ready dataset.
 - Developed ARIMA and Random Forest forecasting models in R and Python, engineering lagged features and covariates to capture both linear and nonlinear drivers of cocoa price volatility.
 - Conducted full model diagnostics, including ACF/PACF analysis, Ljung-Box tests, residual evaluation, and feature importance assessment, to validate model assumptions and benchmark performance.
 - Authored a comprehensive statistical report synthesizing findings, model limitations, and next-step recommendations, highlighting the impact of climate and economic factors on commodity pricing.

Health Studies Course Projects

- Feb 2024 – **Effectiveness of Usage of Virtual Reality in Raising Awareness about Cannabis Misuse Issues among Teenages**, HMB342
Apr 2024
- Designed a randomized controlled trial study comparing virtual reality-based education with traditional seminars to evaluate effectiveness in raising awareness of cannabis misuse among Canadian high school students.
 - Developed evaluation metrics and statistical analysis plan (pre/post surveys, T-tests, knowledge retention, behavioral intentions) to measure intervention impact on awareness and future cannabis use.
 - Applied health promotion and program design frameworks to integrate emerging technologies into adolescent substance misuse education, highlighting cost-effectiveness and policy implications for public health strategies.
- Feb 2023 – **Food Insecurity Intervention Program Planning**, HST330
Apr 2023
- Conducted root cause analysis and developed a program proposal to address food insecurity among Toronto low-income families, modeling intervention design on the Supplemental Nutrition Assistance Program with a digital e-coupon system.
 - Designed a quasi-experimental evaluation plan, including baseline and post-intervention measures (Food Insecurity Experience Scale, BMI, Healthy Eating Index, stress levels) to assess program effectiveness.
 - Created a logic model and theory of change framework to outline expected outcomes, health implications, and long-term policy applications, demonstrating strong skills in program evaluation and health policy analysis.

Oct 2022 – **Proposal to Address Teenager Substance Misuse Problem, HST209**

- Dec 2022
- Conducted comprehensive analysis of risk factors (family influence, peer pressure, media exposure, mental health) contributing to teenage substance misuse, synthesizing evidence-based research into actionable insights.
 - Designed and evaluated multi-level interventions including individual counseling, family therapy, in-school workshops, and peer-led support communities to promote adolescent health and reduce substance misuse.
 - Developed health promotion strategies grounded in empowerment and community engagement, aligning interventions with public health models to improve long-term outcomes for at-risk youth.

Other Experiences

Sep 2023 – **Teaching Assistant and Learning Support, University of Toronto Department of Statistical Sciences and Data Sciences Institute**

DoSS Courses: STA130, STA238, STA304, Statistics Aid Centre, Florence Nightingale Day (2024, 2025)

DSI Modules: SQL, Applying Statistical Concepts, Sampling, Visualization

- Prepared tutorials and office hours for statistics courses, strengthening programming (R, Python, SQL) skills and addressed questions across a wide range of statistical courses, enhancing students' statistical learning.
- Engaged with high school students at events such as Florence Nightingale Day, translating high-level statistical concepts into accessible and engaging explanations to spark interest in statistics.

Sep 2022 – **First-year Learning Community Assistant Peer Mentor, University of Toronto**

- Apr 2023
- Facilitated 12 sessions to guide 20+ first year students in academic matters, fostering their skills to overcome challenges and manage transition anxiety to university life.
 - Established a rapport and inclusive community among students, promoting a supportive learning environment.
 - Coordinated discussions with Mathematics professors and library staff and managed online Quercus page to facilitate remote communication and engagement with students.

May 2022 **Bilingual Client Care Representative, Telus Health**

- Sep 2022
- Served as the first point of contact for 100+ daily client phone call and message inquiries, delivering bilingual (English, Chinese) support that consistently improved satisfaction and reduced call escalations.
 - Assessed client needs, identified the purpose of each inquiry, and provided tailored service recommendations, ensuring clients felt informed, supported, and guided toward the right solutions.
 - Coordinated service delivery by booking appointments and routing callers to the appropriate service-specific teams, contributing to faster resolution times and smoother operations.

Volunteer

Sep 2022 – **Data Entry Volunteer, St. Michael's Hospital Trauma Research Program and Injury Prevention**

- Dec 2022
- Reviewed eligible medical charts to collect injury/poisoning information
 - Documented medical records into the system